
IMAGE-IDENTIFIABLE GENOMIC SUBSPACES: A LINEAR-GAUSSIAN MODEL OF RADIOGENOMIC RECOVERABILITY

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ABSTRACT

We develop a linear-Gaussian generative model linking patient-level genomic profiles $g \in \mathbb{R}^p$ to imaging phenotypes $x \in \mathbb{R}^d$, and use it to ask a sharper question than “does imaging predict gene j ?” Instead we ask *which low-dimensional genomic subspaces are image-identifiable*. Under the model every quantity of interest is available in closed form: the Bayes-optimal estimator $\mathbb{E}[g \mid x]$, the posterior covariance $\Sigma_{G|X}$, the mutual information $I(g; x)$, and a per-direction *recoverability score* $R(v) \in [0, 1]$. We show that the directions of maximal recoverability solve a generalized eigenproblem that coincides exactly with canonical correlation analysis (CCA) between the two modalities, and that the recoverability of the i -th canonical genomic direction equals the squared canonical correlation ρ_i^2 . We then address estimation in the regime $p \gg n$ via regularized covariances, reduced-rank regression, and the probabilistic-CCA latent-variable form, and describe cross-validated and permutation-based procedures for honest inference. The companion notebook `notebooks/radiogenomic_recoverability.ipynb` implements the full pipeline on a pair of `AnnData` objects, with a synthetic-data fallback that has known ground truth.

Keywords radiogenomics · canonical correlation analysis · mutual information · recoverability · linear-Gaussian model

1 Setup and notation

We observe n patients. For patient i let $g_i \in \mathbb{R}^p$ be a vector of genomic features (e.g. expression, somatic mutation burden, copy-number scores) and $x_i \in \mathbb{R}^d$ a vector of imaging phenotypes (radiomic descriptors, deep features, or structured reads) [1–3]. Stack the observations into data matrices $\mathbf{G} \in \mathbb{R}^{n \times p}$ and $\mathbf{X} \in \mathbb{R}^{n \times d}$, with row i equal to $g_i^\top$ and $x_i^\top$ respectively. Random variables are written in uppercase (G, X), realizations in lowercase (g, x). Throughout we assume both modalities have been centered, so all means are zero; recentring is handled by the estimators of Section 10.

We use $\Sigma_G \in \mathbb{R}^{p \times p}$, $\Sigma_X \in \mathbb{R}^{d \times d}$ for the marginal covariances, $\Sigma_{GX} = \text{Cov}(G, X) \in \mathbb{R}^{p \times d}$ for the cross-covariance, and $\Sigma_{XG} = \Sigma_{GX}^\top$. For a symmetric positive semidefinite S we write $S^{1/2}$ for its symmetric square root and $S^{-1/2}$ for the inverse thereof.

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2 The linear–Gaussian generative model

The genomic state is drawn from a Gaussian prior and the imaging phenotype is a noisy linear readout of it:

$$G \sim \mathcal{N}(0, \Sigma_G), \quad (1)$$

$$X | G \sim \mathcal{N}(AG, \sigma^2 I_d), \quad A \in \mathbb{R}^{d \times p}. \quad (2)$$

The matrix A encodes how genomic variation is expressed in the image; the isotropic term $\sigma^2 I_d$ collects everything in the image that genomics does not drive — biological variation outside the measured panel, acquisition and reconstruction noise, and feature-extraction error. Equivalently,

$$X = AG + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d), \quad \varepsilon \perp G. \quad (3)$$

Equations (1)–(3) are deliberately the simplest model with a non-trivial answer. Their value is that *every* downstream quantity is closed-form, so the model functions as an exact upper bound: no estimator — linear or nonlinear — can recover genomics from imaging better than the posterior derived below (under these assumptions). This fundamental-limit framing is in the spirit of rate–distortion theory and the information bottleneck [4–8], specialized here to a tractable linear–Gaussian channel. Section 10 relaxes the isotropic-noise and known-parameter assumptions for practical fitting.

3 Marginal and joint laws

Proposition 1 (Joint Gaussian law). *Under (1)–(3),*

$$\begin{pmatrix} G \\ X \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \Sigma_G & \Sigma_G A^\top \\ A \Sigma_G & A \Sigma_G A^\top + \sigma^2 I_d \end{pmatrix}\right). \quad (4)$$

In particular,

$$X \sim \mathcal{N}(0, \Sigma_X), \quad \Sigma_X = A \Sigma_G A^\top + \sigma^2 I_d, \quad (5)$$

and the cross-covariance is

$$\Sigma_{GX} = \text{Cov}(G, X) = \Sigma_G A^\top. \quad (6)$$

Proof. Since

$$X = AG + \varepsilon,$$

where

$$G \sim \mathcal{N}(0, \Sigma_G), \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d),$$

and G and ε are independent, the stacked vector

$$\begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} G \\ AG + \varepsilon \end{pmatrix}$$

is jointly Gaussian, as it is an affine transformation of independent Gaussian variables.

It therefore suffices to compute its mean and covariance. Since both G and ε are centered,

$$\mathbb{E}[G] = 0, \quad \mathbb{E}[\varepsilon] = 0,$$

we obtain

$$\mathbb{E}[X] = A\mathbb{E}[G] + \mathbb{E}[\varepsilon] = 0,$$

and hence

$$\mathbb{E}\begin{pmatrix} G \\ X \end{pmatrix} = 0.$$

Next, the covariance matrix has block form

$$\text{Cov}\begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} \text{Cov}(G, G) & \text{Cov}(G, X) \\ \text{Cov}(X, G) & \text{Cov}(X, X) \end{pmatrix}.$$

The upper-left block is simply

$$\text{Cov}(G, G) = \Sigma_G.$$

For the upper-right block, using $X = AG + \varepsilon$,

$$\text{Cov}(G, X) = \text{Cov}(G, AG + \varepsilon).$$

By linearity of covariance,

$$\text{Cov}(G, X) = \text{Cov}(G, AG) + \text{Cov}(G, \varepsilon).$$

Since G and ε are independent,

$$\text{Cov}(G, \varepsilon) = 0,$$

and therefore

$$\text{Cov}(G, X) = \text{Cov}(G, AG).$$

Using the identity

$$\text{Cov}(U, BV) = \text{Cov}(U, V)B^\top,$$

we obtain

$$\text{Cov}(G, AG) = \Sigma_G A^\top.$$

Hence

$$\Sigma_{GX} = \text{Cov}(G, X) = \Sigma_G A^\top.$$

The lower-left block is the transpose:

$$\text{Cov}(X, G) = \text{Cov}(G, X)^\top = A\Sigma_G.$$

Finally, for the lower-right block,

$$\text{Cov}(X, X) = \text{Cov}(AG + \varepsilon).$$

Expanding the covariance,

$$\text{Cov}(X, X) = \text{Cov}(AG) + \text{Cov}(\varepsilon),$$

since the cross terms vanish by independence. Using

$$\text{Cov}(AG) = A\Sigma_G A^\top,$$

together with

$$\text{Cov}(\varepsilon) = \sigma^2 I_d,$$

gives

$$\text{Cov}(X, X) = A\Sigma_G A^\top + \sigma^2 I_d.$$

Therefore

$$\Sigma_X = A\Sigma_G A^\top + \sigma^2 I_d.$$

Collecting the covariance blocks yields

$$\text{Cov} \begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} \Sigma_G & \Sigma_G A^\top \\ A\Sigma_G & A\Sigma_G A^\top + \sigma^2 I_d \end{pmatrix},$$

which proves the claim. $\square$

Equation (5) is the model's first interpretive statement: imaging covariance decomposes additively into a *genomics-driven* component $A\Sigma_G A^\top$ and an *irreducible* component $\sigma^2 I_d$. The cross-covariance $\Sigma_{GX} = \Sigma_G A^\top$ is the central estimable object — it names which genomic directions co-vary with imaging at all.

4 Bayes-optimal recovery of genomics from imaging

Proposition 2 (Posterior). *The posterior of genomics given imaging is Gaussian, $G \mid X \sim \mathcal{N}(\hat{g}(X), \Sigma_{G|X})$, with*

$$\hat{g}(X) = \mathbb{E}[G \mid X] = \Sigma_G A^\top (A\Sigma_G A^\top + \sigma^2 I_d)^{-1} X = BX, \quad (7)$$

$$\Sigma_{G|X} = \Sigma_G - \Sigma_G A^\top (A\Sigma_G A^\top + \sigma^2 I_d)^{-1} A\Sigma_G. \quad (8)$$

Equivalently, in precision form,

$$\Sigma_{G|X}^{-1} = \Sigma_G^{-1} + \sigma^{-2} A^\top A, \quad B = \sigma^{-2} \Sigma_{G|X} A^\top. \quad (9)$$

Proof. We start from the joint law of Proposition 1. For a jointly Gaussian pair with zero mean and blocks

$$\begin{pmatrix} G \\ X \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \Sigma_G & \Sigma_{GX} \\ \Sigma_{XG} & \Sigma_X \end{pmatrix}\right),$$

the standard Gaussian conditioning formulas state that $G \mid X$ is Gaussian with

$$\mathbb{E}[G \mid X] = \Sigma_{GX} \Sigma_X^{-1} X, \quad \text{Cov}(G \mid X) = \Sigma_G - \Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}.$$

Substituting the blocks computed in Proposition 1, namely $\Sigma_{GX} = \Sigma_G A^\top$, $\Sigma_{XG} = A \Sigma_G$, and $\Sigma_X = A \Sigma_G A^\top + \sigma^2 I_d$, gives

$$\hat{g}(X) = \mathbb{E}[G \mid X] = \Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I_d)^{-1} X,$$

$$\Sigma_{G \mid X} = \Sigma_G - \Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I_d)^{-1} A \Sigma_G,$$

which are (7)–(8), and identifies the Bayes map $B = \Sigma_G A^\top \Sigma_X^{-1}$.

For the precision form, apply the Woodbury matrix identity

$$(\Sigma_G^{-1} + \sigma^{-2} A^\top A)^{-1} = \Sigma_G - \Sigma_G A^\top (\sigma^2 I_d + A \Sigma_G A^\top)^{-1} A \Sigma_G.$$

The right-hand side is exactly $\Sigma_{G \mid X}$, so $\Sigma_{G \mid X}^{-1} = \Sigma_G^{-1} + \sigma^{-2} A^\top A$; imaging contributes additively to the prior precision Σ_G^{-1} .

It remains to rewrite B . Multiply the push-through identity

$$A^\top (A \Sigma_G A^\top + \sigma^2 I_d) = (A^\top A + \sigma^2 \Sigma_G^{-1}) \Sigma_G A^\top = \sigma^2 (\Sigma_G^{-1} + \sigma^{-2} A^\top A) \Sigma_G A^\top$$

on the right by $(A \Sigma_G A^\top + \sigma^2 I_d)^{-1}$ and on the left by $\sigma^{-2} (\Sigma_G^{-1} + \sigma^{-2} A^\top A)^{-1} = \sigma^{-2} \Sigma_{G \mid X}$ to obtain

$$\sigma^{-2} \Sigma_{G \mid X} A^\top = \Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I_d)^{-1} = B.$$

Hence $B = \sigma^{-2} \Sigma_{G \mid X} A^\top$, completing (9). $\square$

The estimator $\hat{g}(X) = BX$ is the Bayes-optimal predictor under squared-error loss; because the model is jointly Gaussian it is also the optimal predictor over *all* measurable functions of X , not merely linear ones. Thus $\Sigma_{G \mid X}$ is a hard floor on residual genomic uncertainty: **no algorithm can do better**. The precision form (9) makes the accounting transparent — imaging *adds* information $\sigma^{-2} A^\top A$ to the prior precision Σ_G^{-1} .

5 Mutual information and the SNR decomposition

Proposition 3 (Mutual information). *The mutual information between the two modalities is*

$$I(G; X) = \frac{1}{2} \log \frac{\det \Sigma_G}{\det \Sigma_{G \mid X}} = \frac{1}{2} \log \det (I_d + \sigma^{-2} A \Sigma_G A^\top). \quad (10)$$

Let $\lambda_1 \geq \dots \geq \lambda_r > 0$ be the nonzero eigenvalues of $A \Sigma_G A^\top$. Then

$$I(G; X) = \frac{1}{2} \sum_{i=1}^r \log \left(1 + \frac{\lambda_i}{\sigma^2} \right). \quad (11)$$

Proof. For a jointly Gaussian pair the mutual information is the expected log-ratio of the posterior to the prior density of G ; carrying out the Gaussian integral leaves only the determinants of the two covariances,

$$I(G; X) = \frac{1}{2} \log \frac{\det \Sigma_G}{\det \Sigma_{G \mid X}}$$

[9]. We now express the ratio through the channel parameters. Taking determinants in the precision form $\Sigma_{G \mid X}^{-1} = \Sigma_G^{-1} + \sigma^{-2} A^\top A$ of (9) and factoring out Σ_G^{-1} ,

$$\det \Sigma_{G \mid X}^{-1} = \det(\Sigma_G^{-1}) \det(I_p + \sigma^{-2} \Sigma_G^{1/2} A^\top A \Sigma_G^{1/2}),$$

so that

$$\frac{\det \Sigma_G}{\det \Sigma_{G|X}} = \det \Sigma_G \cdot \det \Sigma_{G|X}^{-1} = \det(I_p + \sigma^{-2} \Sigma_G^{1/2} A^\top A \Sigma_G^{1/2}).$$

Sylvester’s determinant identity $\det(I_p + PQ) = \det(I_d + QP)$, applied with $P = \sigma^{-2} \Sigma_G^{1/2} A^\top$ and $Q = A \Sigma_G^{1/2}$, moves the factors past one another and replaces the $p \times p$ determinant by a $d \times d$ one,

$$\det(I_p + \sigma^{-2} \Sigma_G^{1/2} A^\top A \Sigma_G^{1/2}) = \det(I_d + \sigma^{-2} A \Sigma_G A^\top),$$

which is (10). Finally, let $A \Sigma_G A^\top = \sum_{i=1}^r \lambda_i w_i w_i^\top$ be the spectral decomposition of this symmetric positive-semidefinite matrix, with $\lambda_1 \geq \dots \geq \lambda_r > 0$ its nonzero eigenvalues. Then $I_d + \sigma^{-2} A \Sigma_G A^\top$ has eigenvalues $1 + \lambda_i / \sigma^2$ (and 1 on the orthogonal complement, contributing nothing), so its determinant is the product $\prod_{i=1}^r (1 + \lambda_i / \sigma^2)$ and

$$I(G; X) = \frac{1}{2} \log \prod_{i=1}^r \left(1 + \frac{\lambda_i}{\sigma^2}\right) = \frac{1}{2} \sum_{i=1}^r \log \left(1 + \frac{\lambda_i}{\sigma^2}\right),$$

which is (11). □

Equation (11) is the cleanest reading of the model: each image-visible genomic direction contributes $\frac{1}{2} \log(1 + \text{SNR}_i)$ bits (nats), where $\text{SNR}_i = \lambda_i / \sigma^2$ is the signal-to-noise ratio of that direction. Directions whose genomic signal is small relative to imaging noise contribute essentially nothing, regardless of how many genes load on them. Figure 1 illustrates the decomposition (11) on the synthetic instance introduced below: the per-direction bit contribution falls off rapidly after the first handful of canonical components, and the cumulative total saturates at $I(G; X) \approx 6.45$ bits out of the analytic upper bound implied by the model.

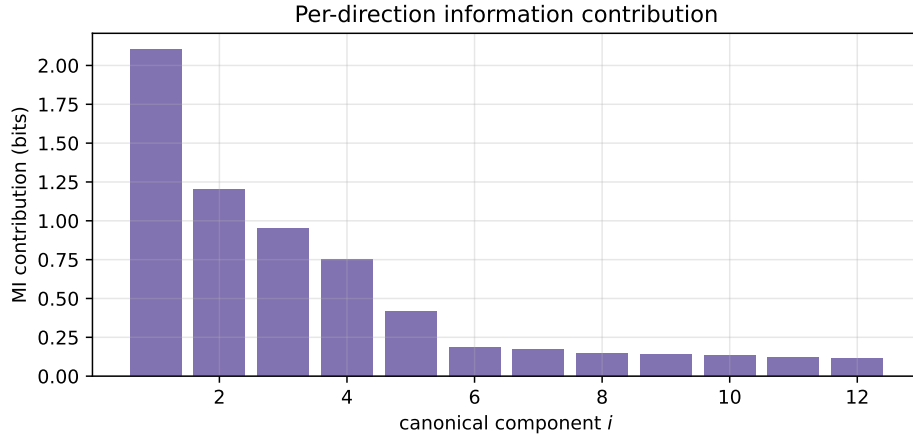


Figure 1: Per-direction contribution to $I(G; X)$ on a synthetic linear–Gaussian instance ($n = 800$, $p = 80$, $d = 40$). Each bar is $\frac{1}{2} \log(1 + d_i^2)$ in bits; the dashed line is the cumulative total.

6 Canonical coordinates and the recoverability spectrum

The eigenvalues λ_i are most interpretable after whitening. Define the whitened genomic variable $U = \Sigma_G^{-1/2} G \sim \mathcal{N}(0, I_p)$ and the *whitened transfer operator*

$$M := \sigma^{-1} A \Sigma_G^{1/2} \in \mathbb{R}^{d \times p}, \quad M = \sum_i d_i u_i v_i^\top \quad (\text{SVD}), \quad (12)$$

with left/right singular vectors $u_i \in \mathbb{R}^d$, $v_i \in \mathbb{R}^p$ and singular values $d_1 \geq d_2 \geq \dots \geq 0$.

Proposition 4 (Canonical decomposition of recoverability). *With M as in (12), define the genomic canonical direction $a_i := \Sigma_G^{-1/2} v_i$ and the canonical score $S_i := a_i^\top G = v_i^\top U$. Then:*

- (i) S_i has prior variance 1 and posterior variance $\text{Var}(S_i | X) = (1 + d_i^2)^{-1}$;

- (ii) the canonical correlation between S_i and imaging is $\rho_i = d_i / \sqrt{1 + d_i^2}$, equivalently $d_i^2 = \rho_i^2 / (1 - \rho_i^2)$;
 (iii) the eigenvalues of Section 5 satisfy $\lambda_i / \sigma^2 = d_i^2$, so

$$I(G; X) = \frac{1}{2} \sum_i \log(1 + d_i^2) = -\frac{1}{2} \sum_i \log(1 - \rho_i^2). \quad (13)$$

Proof. Write the SVD as $M = UDV^\top$ with $U^\top U = I$, $V^\top V = I$, and $D = \text{diag}(d_1, d_2, \dots)$, so $V = [v_1, v_2, \dots]$ collects the right singular vectors and u_i the left ones. Since $S_i = v_i^\top U$ with $U \sim \mathcal{N}(0, I_p)$, the prior variance is $\text{Var}(S_i) = v_i^\top v_i = 1$.

Posterior variance (i). In whitened coordinates the mean readout is $AG = A\Sigma_G^{1/2}U = \sigma MU$, so the likelihood is $X | U \sim \mathcal{N}(\sigma MU, \sigma^2 I_d)$. Applying the precision form (9) to the pair (U, X) — whose prior covariance is I_p and whose transfer matrix is σM with noise $\sigma^2 I_d$ — the posterior precision of U is

$$\text{Var}(U | X)^{-1} = I_p + \sigma^{-2}(\sigma M)^\top (\sigma M) = I_p + M^\top M = V(I_p + D^2)V^\top,$$

using $M^\top M = VD^2V^\top$. Inverting and applying $V^\top V = I_p$,

$$\text{Var}(U | X) = V(I_p + D^2)^{-1}V^\top,$$

and therefore

$$\text{Var}(S_i | X) = v_i^\top \text{Var}(U | X) v_i = [(I_p + D^2)^{-1}]_{ii} = \frac{1}{1 + d_i^2},$$

since the v_i are orthonormal. This proves (i).

Canonical correlation (ii). The cross-covariance of the whitened genomics with imaging is

$$\text{Cov}(U, X) = \mathbb{E}[U(\sigma MU + \varepsilon)^\top] = \sigma \mathbb{E}[UU^\top] M^\top + \mathbb{E}[U\varepsilon^\top] = \sigma M^\top,$$

using $\mathbb{E}[UU^\top] = I_p$ and $U \perp \varepsilon$. Hence $\text{Cov}(S_i, X) = v_i^\top \text{Cov}(U, X) = \sigma v_i^\top M^\top = \sigma d_i u_i^\top$, the last step because $v_i^\top M^\top = (Mv_i)^\top = (d_i u_i)^\top$ from the SVD. Pair S_i with the matched imaging variate $T_i := u_i^\top X$. From $\Sigma_X = \sigma^2(MM^\top + I_d)$ (Proposition 1 in whitened form) and $MM^\top u_i = d_i^2 u_i$,

$$\text{Var}(T_i) = u_i^\top \Sigma_X u_i = \sigma^2(u_i^\top MM^\top u_i + 1) = \sigma^2(1 + d_i^2),$$

while $\text{Cov}(S_i, T_i) = \text{Cov}(S_i, X) u_i = \sigma d_i u_i^\top u_i = \sigma d_i$. The correlation is therefore

$$\rho_i = \text{corr}(S_i, T_i) = \frac{\text{Cov}(S_i, T_i)}{\sqrt{\text{Var}(S_i) \text{Var}(T_i)}} = \frac{\sigma d_i}{\sqrt{1} \sqrt{\sigma^2(1 + d_i^2)}} = \frac{d_i}{\sqrt{1 + d_i^2}},$$

and rearranging gives $d_i^2 = \rho_i^2 / (1 - \rho_i^2)$, proving (ii).

Connection to the eigenvalues (iii). From $M = \sigma^{-1} A \Sigma_G^{1/2}$ we get $A \Sigma_G A^\top = \sigma^2 M M^\top$, whose nonzero eigenvalues are $\sigma^2 d_i^2$; comparing with the λ_i of Proposition 3 gives $\lambda_i = \sigma^2 d_i^2$, i.e. $\lambda_i / \sigma^2 = d_i^2$. Substituting into (11) and using $1 + d_i^2 = 1 / (1 - \rho_i^2)$ from (ii),

$$I(G; X) = \frac{1}{2} \sum_i \log(1 + d_i^2) = -\frac{1}{2} \sum_i \log(1 - \rho_i^2),$$

which is (iii). □

Part (i) gives the headline identity. Define the recoverability of a genomic score as one minus the fraction of its variance that survives observing imaging (formalized in Section 7); then

$$R(a_i) = 1 - \text{Var}(S_i | X) = 1 - \frac{1}{1 + d_i^2} = \frac{d_i^2}{1 + d_i^2} = \rho_i^2. \quad (14)$$

The recoverability of the i -th canonical genomic direction is exactly the squared canonical correlation. This is what reduces the whole analysis to CCA between $\mathbf{G}$ and $\mathbf{X}$ (Section 10), and it means A and σ^2 never have to be estimated individually — only the three second moments $\Sigma_G, \Sigma_X, \Sigma_{GX}$ enter.

7 Recoverability of a named genomic direction

In practice one often cares about a specific, biologically named genomic score $Y = v^\top G$ for a fixed $v \in \mathbb{R}^p$ — a pathway signature, a polygenic score, a single gene's expression ($v = e_j$).

Definition 1 (Recoverability score). For $v \in \mathbb{R}^p$ with $v^\top \Sigma_G v > 0$,

$$R(v) := 1 - \frac{v^\top \Sigma_{G|X} v}{v^\top \Sigma_G v} = \frac{v^\top (\Sigma_G - \Sigma_{G|X}) v}{v^\top \Sigma_G v} \in [0, 1]. \quad (15)$$

$R(v) = 0$ means imaging carries no information about $Y = v^\top G$; $R(v) = 1$ means imaging determines it exactly. The score is scale-invariant in v and, by the Gaussian conditional-variance identity, equals the population R^2 of the Bayes-optimal predictor of Y from X : $R(v) = 1 - \mathbb{E}[\text{Var}(Y | X)] / \text{Var}(Y)$. The matrix at the core of (15) has the convenient moment-only form

$$\Sigma_G - \Sigma_{G|X} = \Sigma_G A^\top \Sigma_X^{-1} A \Sigma_G = \Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}. \quad (16)$$

8 The recoverability eigenproblem

The natural project-level question — *which* genomic directions does imaging see best — is the variational problem

$$\max_{v \neq 0} R(v) = \max_{v \neq 0} \frac{v^\top (\Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}) v}{v^\top \Sigma_G v}. \quad (17)$$

Proposition 5 (Recoverability eigenproblem = CCA). *The stationary points of (17) are the solutions of the generalized eigenproblem*

$$\Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG} v = \mu \Sigma_G v. \quad (18)$$

Its eigenvalues are $\mu_i = \rho_i^2 = d_i^2 / (1 + d_i^2)$ and its eigenvectors are the canonical genomic directions a_i of Proposition 4, Σ_G -orthonormal ($a_i^\top \Sigma_G a_j = \delta_{ij}$). Thus the ordered solutions of (18) are precisely the canonical correlation directions between G and X , and $\max_v R(v) = \rho_1^2$.

Proof. Write $C := \Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}$ for the (symmetric positive-semidefinite) numerator matrix, so the objective is the generalized Rayleigh quotient $R(v) = v^\top C v / v^\top \Sigma_G v$. Setting its gradient to zero,

$$\nabla_v R(v) = \frac{2Cv (v^\top \Sigma_G v) - 2\Sigma_G v (v^\top C v)}{(v^\top \Sigma_G v)^2} = 0 \iff Cv = \frac{v^\top C v}{v^\top \Sigma_G v} \Sigma_G v = \mu \Sigma_G v,$$

with $\mu = R(v)$ the value of the quotient at the stationary point. This is the generalized eigenproblem (18).

To solve it, whiten via $v = \Sigma_G^{-1/2} w$. Left-multiplying (18) by $\Sigma_G^{-1/2}$ and inserting $\Sigma_G^{-1/2} \Sigma_G^{1/2} = I$ turns it into the ordinary symmetric eigenproblem

$$(\Sigma_G^{-1/2} C \Sigma_G^{-1/2}) w = \mu w.$$

Now expand C . Using $\Sigma_{GX} = \Sigma_G A^\top$ and $\Sigma_{XG} = A \Sigma_G$,

$$\Sigma_G^{-1/2} C \Sigma_G^{-1/2} = \Sigma_G^{-1/2} (\Sigma_G A^\top) \Sigma_X^{-1} (A \Sigma_G) \Sigma_G^{-1/2} = (\Sigma_G^{1/2} A^\top) \Sigma_X^{-1} (A \Sigma_G^{1/2}).$$

Substituting $A \Sigma_G^{1/2} = \sigma M$ and $\Sigma_X = \sigma^2 (M M^\top + I_d)$,

$$\Sigma_G^{1/2} A^\top \Sigma_X^{-1} A \Sigma_G^{1/2} = (\sigma M^\top) \cdot \sigma^{-2} (M M^\top + I_d)^{-1} \cdot (\sigma M) = M^\top (M M^\top + I_d)^{-1} M.$$

Insert the SVD $M = U D V^\top$. Then $M M^\top + I_d = U D^2 U^\top + I_d$, and on the range of U its inverse acts as $U (D^2 + I)^{-1} U^\top$, so

$$M^\top (M M^\top + I_d)^{-1} M = V D U^\top \cdot U (D^2 + I)^{-1} U^\top \cdot U D V^\top = V D^2 (D^2 + I)^{-1} V^\top,$$

using $U^\top U = I$. This is an eigendecomposition: the eigenvalues are $d_i^2 / (1 + d_i^2)$ with eigenvectors $w = v_i$. By Proposition 4(ii) these eigenvalues equal ρ_i^2 , so $\mu_i = \rho_i^2$ and the leading value is $\max_v R(v) = \mu_1 = \rho_1^2$.

Undoing the whitening, the genomic eigenvector is $v = \Sigma_G^{-1/2} w = \Sigma_G^{-1/2} v_i = a_i$, the canonical direction of Proposition 4. Finally, the Σ_G -orthonormality follows because the v_i are orthonormal:

$$a_i^\top \Sigma_G a_j = (\Sigma_G^{-1/2} v_i)^\top \Sigma_G (\Sigma_G^{-1/2} v_j) = v_i^\top v_j = \delta_{ij}.$$

□

So the deliverable of the analysis is a *recoverability spectrum* $\rho_1^2 \geq \rho_2^2 \geq \dots$ together with the genomic directions a_i that achieve it. Reading the loadings of $a_1, a_2, \dots$ back onto genes or pathways answers “which biology is image-identifiable”: the leading directions typically align with coarse, spatially expressed programs — proliferation, hypoxia, angiogenesis, necrosis, immune infiltration, tumor burden, stromal remodeling — rather than with individual transcripts, consistent with clear-cell RCC radiogenomic findings [10–12].

Illustration on synthetic data. Figure 2 shows the recoverability spectrum on a synthetic radiogenomic instance generated from the linear–Gaussian model of Section 2 ($n = 800, p = 80, d = 40$, latent rank $k = 5$ with prescribed $\{\rho_i^*\}$). The estimated $\hat{\rho}_i^2$ closely match the planted values for the identifiable directions and decay toward the high-dimensional bulk thereafter, validating Proposition 5. Figure 3 reads the same estimator against the analytic ground truth derived from Propositions 2–4: the recovered posterior mean $\hat{B}X$, posterior-variance retention, and direction alignment all track the closed-form predictions.

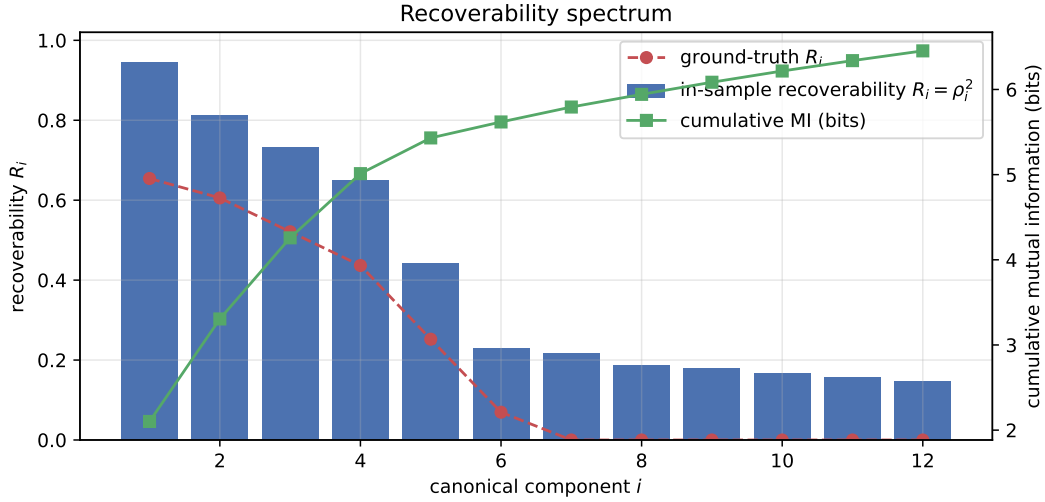


Figure 2: Recoverability spectrum on a synthetic linear–Gaussian instance ($n = 800, p = 80, d = 40$). The leading $\hat{\rho}_i^2$ recovers the planted canonical structure; later components fall to the high-dimensional bulk.

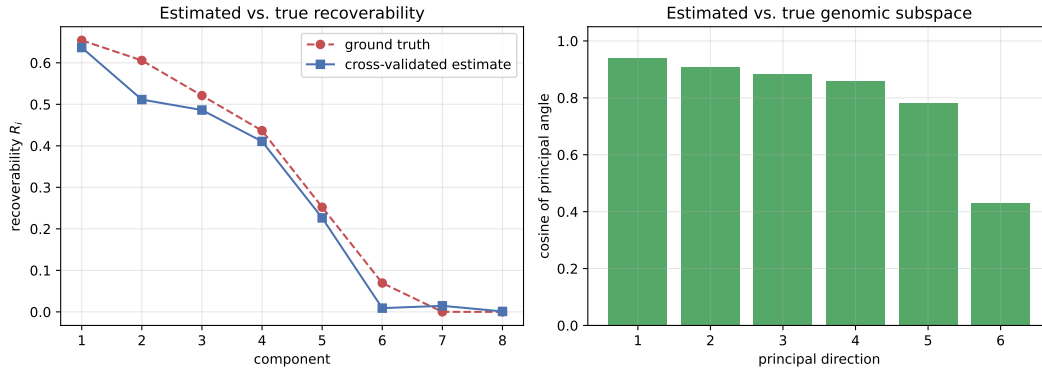


Figure 3: Empirical-vs-analytic check on the same synthetic instance. Estimated canonical correlations, recoverability per direction, and imaging variance attributable to genomics all match the closed-form quantities from Propositions 2–4.

9 Rank limits and identifiability

Proposition 6 (Rank limit). *The recoverable signal matrix has rank*

$$\text{rank}(\Sigma_G - \Sigma_{G|X}) = \text{rank}(A\Sigma_G A^\top) \leq \min(d, p, \text{rank } A, \text{rank } \Sigma_G). \quad (19)$$

Consequently at most $\min(d, p, \text{rank } A, \text{rank } \Sigma_G)$ canonical correlations are nonzero, and the image-identifiable genomic subspace $\text{span}\{a_i : \rho_i > 0\}$ has dimension bounded by the same quantity.

This is the conceptual core. Even with $p = 20,000$ measured genes, imaging can expose only a projection of genomics whose dimension is capped by the rank of the transfer map A — if a CT phenotype reflects, say, 20 macroscopic biological axes, then no model recovers 20,000 independent molecular variables from it. The recoverable dimension is a property of the *biology–imaging channel*, not of the estimator. A data-processing remark: imaging features are themselves a deterministic, lossy function $x = \phi(\text{image})$; any such post-processing can only shrink $\text{rank}(\Sigma_{GX})$ and lower each ρ_i , never raise them, by the data processing inequality [7, 9, 13–15].

10 Estimation from finite samples

In applications p and d may rival or exceed n , the noise is not isotropic, and A, Σ_G, σ^2 are unknown. Because Propositions 4 and 5 require only $\Sigma_G, \Sigma_X, \Sigma_{GX}$, we estimate those directly.

10.1 Regularized covariances

Replace sample covariances by shrinkage estimators (e.g. Ledoit–Wolf [16]), $\hat{\Sigma}_G = (1 - \alpha_G) S_G + \alpha_G \nu_G I_p$ and likewise $\hat{\Sigma}_X$, with $S_G = n^{-1} \mathbf{G}^\top \mathbf{G}$ the centered sample covariance; the cross-covariance is $\hat{\Sigma}_{GX} = n^{-1} \mathbf{G}^\top \mathbf{X}$. Shrinkage keeps $\hat{\Sigma}_G, \hat{\Sigma}_X$ invertible and well-conditioned, which is required for the whitening in (12). When p is very large it is practical to first project genomics onto its top principal components (or onto curated pathway scores), reducing p to a manageable $p' \ll n$ before estimation; the rank limit of Proposition 6 means little is lost provided p' exceeds the recoverable dimension.

Gaussian-copula marginals. The model’s Gaussian prior constrains the *marginals* of each feature, not only their dependence. Radiomic descriptors in particular are badly non-Gaussian — heavy-tailed shape and texture statistics with excess kurtosis in the tens (Section 13) — so the raw second moments $\hat{\Sigma}_G, \hat{\Sigma}_X$ are dominated by a few outlying patients and the whitening in (12) is unstable. A simple and assumption-faithful remedy is the *Gaussian-copula marginal transform*: replace each feature by its rank mapped through the inverse normal CDF, $\bar{M}_{ij} = \Phi^{-1}(\text{rank}_i(M_{.j})/(n+1))$, so every marginal is exactly standard normal and only the dependence structure (hence the canonical correlations) survives. This is the continuous-feature analogue of the latent-Gaussian construction of Section 12 and is the default applied to both modalities in the analyses below.

10.2 Reduced-rank and ridge regression

The transfer map, if needed explicitly, is the multivariate regression of X on G . Ridge gives $\hat{A}^\top = (\mathbf{G}^\top \mathbf{G} + \gamma I_p)^{-1} \mathbf{G}^\top \mathbf{X}$, and $\hat{\sigma}^2$ follows from the residual covariance. Imposing $\text{rank}(A) \leq k$ (reduced-rank regression) bakes Proposition 6 into the fit and is equivalent, up to regularization, to truncating the SVD of M at k components.

10.3 Regularized CCA

The estimator used in the companion notebook solves (18) with the regularized covariances of Section 10.1: form $\hat{M} = \hat{\Sigma}_G^{-1/2} \hat{\Sigma}_{GX} \hat{\Sigma}_X^{-1/2}$, take its SVD $\hat{M} = \hat{U} \hat{D} \hat{V}^\top$, and read off $\hat{\rho}_i = \hat{d}_i$ (clipped to $[0, 1]$), $\hat{R}_i = \hat{\rho}_i^2$, genomic directions $\hat{a}_i = \hat{\Sigma}_G^{-1/2} \hat{u}_i$, imaging directions $\hat{b}_i = \hat{\Sigma}_X^{-1/2} \hat{v}_i$. The Bayes-optimal predictor and posterior covariance follow from (7)–(8) with hats throughout. This is exactly regularized CCA — the recoverability program and CCA are the same computation.

10.4 Probabilistic CCA: the latent-variable view

The rank limit motivates an explicit low-dimensional generative form. With a shared latent $Z \in \mathbb{R}^k$,

$$Z \sim \mathcal{N}(0, I_k), \quad G = W_G Z + \mu_G + \epsilon_G, \quad X = W_X Z + \mu_X + \epsilon_X, \quad (20)$$

$\epsilon_G \sim \mathcal{N}(0, \Psi_G), \epsilon_X \sim \mathcal{N}(0, \Psi_X)$ independent (probabilistic CCA; Bach & Jordan, 2005). Here k is the dimension of the image-identifiable genomic subspace, Z is the shared biological state, and ϵ_G, ϵ_X are modality-private variation. The implied moments are $\Sigma_G = W_G W_G^\top + \Psi_G, \Sigma_X = W_X W_X^\top + \Psi_X, \Sigma_{GX} = W_G W_X^\top$; the maximum-likelihood loadings are the canonical directions scaled by the canonical correlations, recovering Section 10.3. The model is fit by

EM, which also yields posterior latent estimates $\mathbb{E}[Z \mid x]$ and naturally accommodates anisotropic (per-feature) noise, unlike the isotropic $\sigma^2 I_d$ of the base model.

10.5 Sample-size regimes and dimensional collapse

Equations (7)–(18) are population identities; the estimators of Sections 10.1–10.4 inherit a finite-sample bias whose magnitude is governed entirely by the aspect ratios p/n and d/n . The bias is not benign, and below a critical sample size the recoverability spectrum breaks down in a specific, predictable way.

Three regimes. Let p^*, d^* be the working dimensions after any pre-reduction (e.g. PCA), and write $\kappa_G = p^*/n$, $\kappa_X = d^*/n$.

1. *Classical* ($\kappa_G + \kappa_X \ll 1$). Sample CCA is consistent; $\hat{\rho}_i \rightarrow \rho_i$ at the usual $n^{-1/2}$ rate and the in-sample spectrum is a faithful estimate of the population one.
2. *High-dimensional* ($\kappa_G + \kappa_X \lesssim 1$, both bounded away from 1). Under the null $\Sigma_{GX} = 0$, the squared sample canonical correlations follow a Wachter-type law on $[(\sqrt{\kappa_G(1-\kappa_X)} - \sqrt{\kappa_X(1-\kappa_G)})^2, (\sqrt{\kappa_G(1-\kappa_X)} + \sqrt{\kappa_X(1-\kappa_G)})^2]$ [17, 18]; the leading sample value already sits well above zero by geometry. Estimates of ρ_i are systematically inflated and ordinary CCA must be replaced by its regularized form, with shrinkage parameters tuned by cross-validation.
3. *Saturating* ($p^* + d^* \geq n$). The rows of $\mathbf{G}$ and $\mathbf{X}$ live in subspaces whose intersection has dimension at least $p^* + d^* - n + 1$; consequently $\hat{\rho}_1 = \dots = \hat{\rho}_q = 1$ for $q = \min(p^*, d^*) - (n - \max(p^*, d^*))_+$ regardless of any underlying association. The empirical mutual information diverges, the in-sample $\text{tr}(\hat{\Sigma}_{GX} \hat{\Sigma}_X^{-1} \hat{\Sigma}_{GX}) / \text{tr}(\hat{\Sigma}_X)$ approaches the rank fraction of the genomic span, and the permutation null collapses onto the observed value (its p -values concentrate near 1). Nothing in the spectrum is estimable in this regime without shrinkage *plus* dimension reduction; shrinkage alone keeps the matrices invertible but does not restore identifiability.

Quantitative warning signs. Three diagnostics, jointly, separate real signal from dimensional artefact:

- the gap between in-sample $\hat{R}_i$ and the K -fold value $\hat{R}_i^{\text{cv}}$; under regime 3 this gap approaches $\hat{R}_i - 0$;
- the permutation null for $\hat{\rho}_1$: if the observed value lies inside the bulk, the result is consistent with chance;
- the cross-validated R^2 of the leading canonical score $a_1^\top g$ predicted from x on a held-out split; in regime 3 it is typically large and *negative* (worse than predicting the mean), a behaviour seen empirically on TCGA-KIRC at $n = 131$ with $p^* = 100$, $d^* = 107$ in the companion notebook.

Recommendations. Choose p^* and d^* *adaptively from n* , not from the native modality dimensions:

- Target $p^* + d^* \leq n/\tau$ with $\tau \geq 5$ as a working rule (we use $p^*, d^* \leq \lfloor n/\tau \rfloor$ per side in the notebook; $\tau \in [5, 10]$ trades bias for resolution). Reduce both modalities by PCA when needed — reducing only the larger side leaves the estimator in regime 3.
- Treat $\hat{R}_i^{\text{cv}}$ (Section 10.6) as the primary estimand and report the in-sample $\hat{R}_i$ only with the gap made explicit; the in-sample fraction of imaging variance attributable to genomics should be reported alongside a cross-validated counterpart, e.g. $\sum_i \hat{R}_i^{\text{cv}} / \min(p^*, d^*)$.
- Power planning. The leading population correlation ρ_1 is detectable above the Wachter bulk only when its squared value exceeds $(\sqrt{\kappa_G(1-\kappa_X)} + \sqrt{\kappa_X(1-\kappa_G)})^2$; solving for n gives a minimum sample size given a hypothesised effect. For $\rho_1^2 = 0.2$ and a balanced $p^* = d^*$, this requires roughly $n \gtrsim 25 p^*$, an order of magnitude beyond what cohort sizes in radiogenomics usually offer — reinforcing the need for aggressive pre-reduction.
- Probabilistic CCA (Section 10.4) with a small, fixed latent dimension k is the most parsimonious option when n is the binding constraint; rank- k regularization sidesteps the high-dimensional bulk by construction.

The practical message is conservative: in radiogenomic cohorts, sample size is almost always the binding constraint and a defensible analysis is the one whose effective $p^* + d^*$ is a small fraction of n , with both sides reduced symmetrically and every reported number a cross-validated or permutation-calibrated one.

Figure 4 illustrates these regimes on the synthetic instance, where the planted truth is known. As n grows with p^*, d^* scaled to keep $\kappa_G + \kappa_X$ bounded, the in-sample $\hat{\rho}_1$ sits at or near 1 across the small- n saturating regime — indistinguishable from the permutation null — while the cross-validated $\hat{R}_1^{cv}$ only separates from zero once n clears the Wachter edge. The empirical mutual information shows the same threshold behaviour.

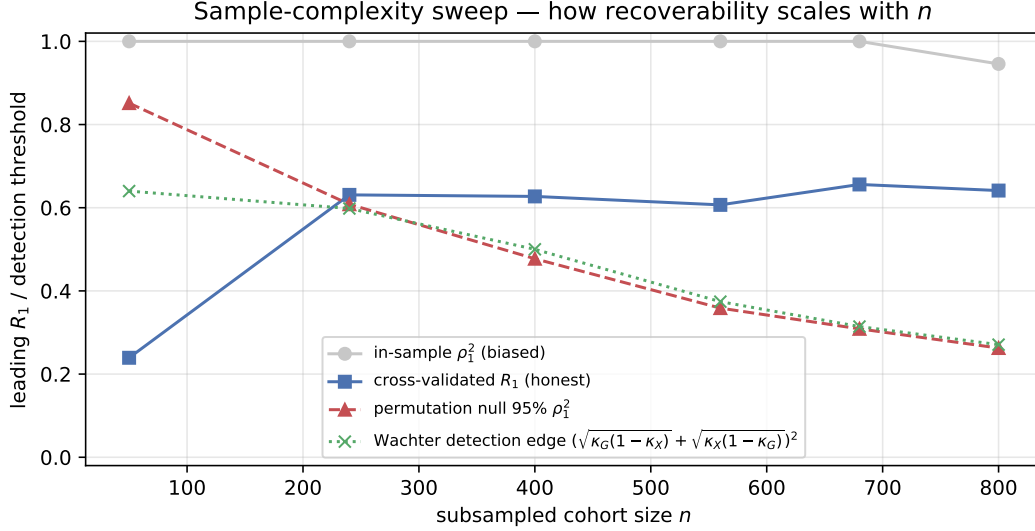


Figure 4: Sample-complexity sweep on the synthetic instance with known truth. In-sample $\hat{\rho}_1$ tracks the null until n exceeds the Wachter edge; cross-validated $\hat{R}_1^{cv}$ and empirical $\hat{I}(G; X)$ only become informative past that threshold. Compare to Figure 10 for the same diagnostic on TCGA-KIRC.

10.6 Honest recoverability: cross-validation and permutation

In-sample canonical correlations are biased upward, severely so when p, d are not small relative to n . Two safeguards, both implemented in the notebook:

- *Cross-validation.* Estimate $\hat{a}_i, \hat{b}_i$ on a training fold; on the held-out fold form the scores $\hat{a}_i^\top g$ and $\hat{b}_i^\top x$ and report their out-of-sample squared correlation as the honest $\hat{R}_i$. The gap to the in-sample value quantifies overfitting.
- *Permutation test.* Repeatedly permute the patient labels of one modality, refit, and collect the leading canonical correlation to build a null distribution; the permutation p -value tests whether any genomic direction is image-identifiable beyond chance.

Scale-matched effective rank. A subtlety arises when these two safeguards are combined to count *how many* directions are identifiable. The in-sample permutation null of the leading $\hat{\rho}_1$ is itself inflated toward 1 by the high-dimensional geometry of Section 10.5, whereas $\hat{R}_i^{cv}$ is honestly deflated; thresholding the deflated statistic against the inflated null forces the estimated rank to zero regardless of any true signal. The two must be compared on the same scale. We therefore build the null of the *cross-validated* statistic directly: permute patient labels, recompute the full K -fold $\hat{R}_i^{cv}$, and define the *effective image-identifiable rank* as the number of directions whose observed $\hat{R}_i^{cv}$ exceeds its own permutation 95% quantile at permutation $p < 0.05$. This is the estimand we report below.

Figure 5 shows both diagnostics on the synthetic instance. The in-sample $\hat{R}_i$ overshoots the planted truth by the amount predicted by the high-dimensional bias, while the 5-fold $\hat{R}_i^{cv}$ tracks the truth for identifiable directions and collapses toward the null 95% quantile thereafter. Figure 6 shows the permutation null on $\hat{\rho}_1$: when the true signal is nonzero, the observed value sits cleanly outside the null bulk and the per-direction p -values fall below 0.01 for the top three components, matching the planted rank.

The same optimism appears in the aggregate fraction of imaging variance attributable to genomics. Figure 7 shows the in-sample share $\sum_i \hat{R}_i / \min(p^*, d^*)$ alongside its cross-validated counterpart on the synthetic instance: the in-sample number sits well above the planted truth while the CV value tracks it. The same diagnostic on TCGA-KIRC

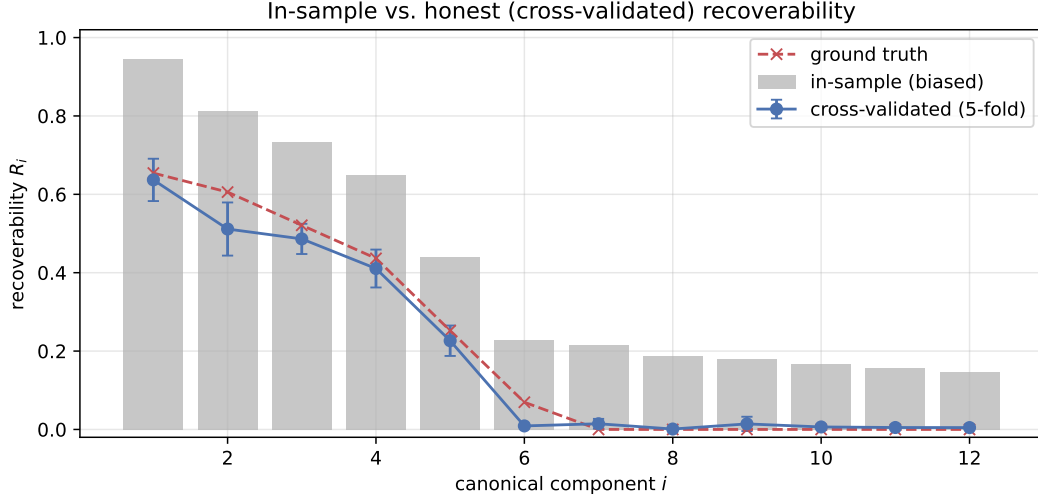


Figure 5: Cross-validated recoverability on a synthetic instance with planted rank $k = 5$. In-sample $\hat{R}_i$ (grey) is biased upward; 5-fold $\hat{R}_i^{cv}$ (blue) tracks the planted truth for the identifiable directions and collapses to the null floor afterwards.

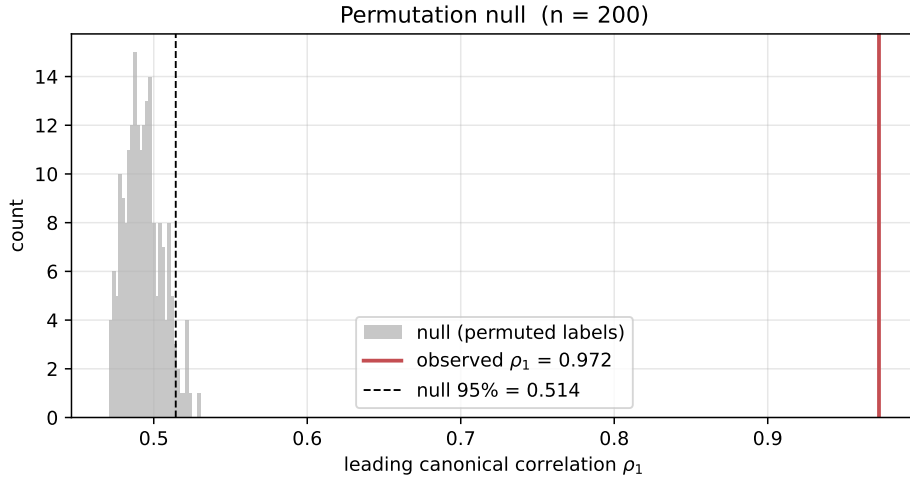


Figure 6: Permutation null on the leading canonical correlation $\hat{\rho}_1$ for the synthetic instance. Red marks the observed value; $p < 0.01$ for the top three directions, recovering the planted rank.

(Section 13) reports 0.193 in-sample against 0.012 cross-validated — a sixteen-fold gap that is consistent with the synthetic mechanism rather than with any genuinely recoverable signal.

11 Interpretation for radiogenomics

The model reframes the radiogenomic question [2, 19–21]. Rather than testing whether CT predicts each gene — a high-dimensional, low-power multiple-testing problem — it asks for the *recoverability spectrum* $\{\rho_i^2\}$ and the associated genomic directions $\{a_i\}$. The outputs are: (1) a small number of image-identifiable genomic axes, with gene/pathway loadings that make them biologically legible; (2) a hard bound $\Sigma_{G|X}$ on what any predictor can achieve, per direction; (3) a total information budget $I(G; X)$; and (4) an honest, cross-validated estimate of each. The governing principle is that radiogenomics should target *image-identifiable low-dimensional genomic subspaces*, whose dimension is set by the biology–imaging channel and not by the gene panel size.

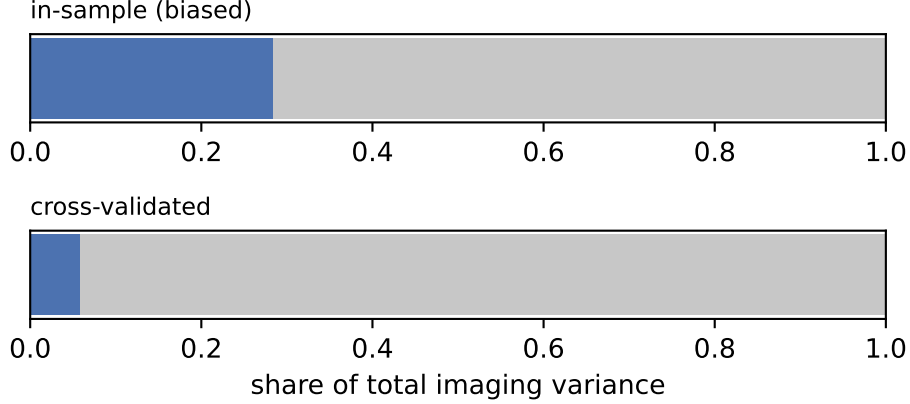


Figure 7: Share of imaging variance attributable to genomics on the synthetic instance: in-sample (grey) versus 5-fold cross-validated (blue), with the planted truth marked. The in-sample share is biased upward by the high-dimensional inflation described in Section 10.5.

12 A Bernoulli extension for mutation data via a Gaussian copula

The linear–Gaussian model of Section 1 is well matched to continuous, approximately log-normal genomic readouts: log-CPM gene expression, methylation β - or M -values, log-CNV ratios. It is plainly misspecified for somatic mutation calls, where each $G_j \in \{0, 1\}$ indicates the presence of a non-silent variant in gene j and the marginal frequencies p_j are small, gene-specific, and highly heterogeneous. This section extends the framework to such data by replacing the Gaussian prior on G with a *Gaussian-copula Bernoulli* (equivalently, multivariate probit) prior, while preserving the canonical-correlation / mutual-information machinery of Sections 5–6.

12.1 The latent-Gaussian generative model

Let $p_j = \Pr(G_j = 1)$ and $\tau_j = \Phi^{-1}(1 - p_j)$, where Φ is the standard normal CDF. Introduce a latent vector

$$Z \sim \mathcal{N}(0, \Sigma_Z), \quad \text{diag}(\Sigma_Z) = I_p,$$

and define the observed mutation indicator by thresholding:

$$G_j = \mathbf{1}\{Z_j > \tau_j\}, \quad j = 1, \dots, p.$$

By construction the marginals are correct, $G_j \sim \text{Bern}(p_j)$, and the pairwise joint laws are governed by the tetrachoric correlation $\Sigma_{Z,jk}$, which can be any positive-definite correlation matrix — providing arbitrary dependence structure between mutated genes (e.g. mutual exclusivity, co-occurrence within pathways).

Imaging is coupled to the *latent* state, not to the indicators directly:

$$X | Z \sim \mathcal{N}(BZ, \sigma^2 I_d), \quad B \in \mathbb{R}^{d \times p}. \quad (21)$$

This is the natural specification: the biological mechanism that produces an imageable phenotype (proliferation, vascularity, necrosis) responds to an underlying continuous burden, of which an observed mutation is a coarse, binarized readout. The joint structure factors as

$$Z \rightarrow G, \quad Z \rightarrow X,$$

so G and X are conditionally independent given Z .

12.2 Mutual information and a tight DPI bound

Because G is a deterministic, memoryless function of Z , the data processing inequality yields the upper bound

$$I(G; X) \leq I(Z; X), \quad (22)$$

with equality only if Z can be recovered from G (impossible whenever any $p_j \in (0, 1)$, since thresholding is lossy). The right-hand side reduces exactly to the Gaussian formula of Section 5: if $\{\rho_i\}_{i=1}^k$ denote the canonical correlations

of (Z, X) under (21), then

$$I(Z; X) = -\frac{1}{2} \sum_{i=1}^k \log(1 - \rho_i^2), \quad k = \text{rank}(\Sigma_Z^{1/2} B^\top \Sigma_X^{-1} B \Sigma_Z^{1/2}), \quad (23)$$

with $\Sigma_X = B \Sigma_Z B^\top + \sigma^2 I_d$. The recoverability spectrum, the directions $\{a_i\}$, and the eigenproblem of Section 8 all transfer verbatim, with Σ_G replaced by Σ_Z .

The binarization gap. The slack in (22) is

$$I(Z; X) - I(G; X) = H(Z | G) - H(Z | G, X) = I(Z; X | G),$$

the residual information about Z not already revealed by the mutation call. For a single gene with frequency p_j , the entropy lost to binarization is $H(Z_j) - H(G_j) = \frac{1}{2} \log(2\pi e) - h_2(p_j)$ where h_2 is the binary entropy in nats; rare mutations ($p_j \ll 1/2$) are highly lossy, common ones near $p_j = 1/2$ less so. Thus (22) is tightest for cohorts dominated by intermediate-frequency mutations and loosest for rare-variant panels — a useful guide to when the bound is informative.

12.3 Per-gene recoverability and AUC ceilings

For a named mutation G_j , the Bayes-optimal predictor from X is the posterior probability

$$\pi_j(x) = \Pr(G_j = 1 | X = x) = \Phi\left(\frac{\mu_j(x) - \tau_j}{s_j}\right),$$

where $(\mu_j(x), s_j^2) = (\mathbb{E}[Z_j | X = x], \text{Var}(Z_j | X = x))$ are the linear-Gaussian posterior moments for the latent coordinate, computed from (Σ_Z, B, σ^2) exactly as in Section 4. The *latent recoverability* $R_j^Z = \text{Var}(\mathbb{E}[Z_j | X]) / \text{Var}(Z_j)$ is the direct analogue of $R(v)$ in the Gaussian case, and the binary AUC of any predictor $\hat{G}_j(x)$ is bounded by

$$\text{AUC}_j \leq \Phi\left(\sqrt{\frac{R_j^Z}{1 - R_j^Z}} / \sqrt{2}\right), \quad (24)$$

attained by the Bayes classifier $\pi_j(x) > p_j$. Equation (24) is the natural object to compare against the empirical per-gene AUC_{LR} and AUC_{RF} of Section 13.6: an empirical AUC near 0.5 is consistent with either a small R_j^Z or a degenerate p_j , and (24) disentangles the two.

12.4 Estimation

The estimator splits into three independent pieces, each well-studied:

- (i) **Marginal thresholds.** Estimate $\hat{p}_j = \bar{G}_{\cdot j}$ and set $\hat{\tau}_j = \Phi^{-1}(1 - \hat{p}_j)$. Apply a continuity correction or empirical-Bayes shrinkage for rare mutations.
- (ii) **Latent correlation** $\hat{\Sigma}_Z$. Estimate pairwise tetrachoric correlations from 2×2 tables of (G_j, G_k) by maximum likelihood under the bivariate-normal copula, then project the resulting matrix to the nearest positive-definite correlation matrix (e.g. Higham’s `nearcorr` [22]). For high-dimensional panels apply Ledoit–Wolf shrinkage as in Section 10.1.
- (iii) **Cross-modal covariance** $\hat{\Sigma}_{ZX}$. The point-biserial covariance $\text{Cov}(G_j, X_\ell)$ is related to the latent covariance by

$$\text{Cov}(G_j, X_\ell) = \phi(\tau_j) \text{Cov}(Z_j, X_\ell),$$

where ϕ is the standard normal density. Estimate $\text{Cov}(G_j, X_\ell)$ from the sample and rescale by $\phi(\hat{\tau}_j)$ to obtain $\hat{\Sigma}_{ZX, j\ell}$.

With $(\hat{\Sigma}_Z, \hat{\Sigma}_{ZX}, \hat{\Sigma}_X)$ in hand, the regularized CCA of Section 10.3, the permutation null of Section 10.6, and the empirical MI bound (estimated, e.g., by k -nearest-neighbour or neural estimators [23–25]) carry over without further change. The dimensional-collapse warnings of Section 10.5 are if anything *more* acute, since tetrachoric estimation has higher variance than Pearson correlation, particularly for rare-rare gene pairs.

12.5 Mixed-modality genomics

In practice a single cohort carries both continuous (expression, CNV) and binary (mutation) features. The copula formulation accommodates this by partitioning the latent vector as $Z = (Z^{\text{cts}}, Z^{\text{bin}})$ with a single joint Σ_Z . The continuous block uses the identity link ($G^{\text{cts}} = Z^{\text{cts}}$ up to marginal standardization) and the binary block uses the probit link of Section 12.1. Estimation of Σ_Z proceeds blockwise — Pearson within the continuous block, tetrachoric within the binary block, polyserial across blocks — after which the rest of the pipeline is identical. This is the route by which the present manuscript’s gene-expression analysis (Section 13) and the per-gene mutation validation (Section 13.6) can be unified into a single coherent recoverability calculation.

13 Application: TCGA-KIRC

We now instantiate the framework on two real radiogenomic cohorts — TCGA-KIRC (clear-cell renal cell carcinoma) [26] and a non-small-cell lung cancer (NSCLC) cohort [27] — to demonstrate how the recoverability spectrum, the cross-validated and permutation-calibrated diagnostics of Section 10.6, and the dimensional-collapse warnings of Section 10.5 interact on typical cohorts. The deliverable is not a predictor but a calibrated answer to “how many genomic directions does imaging see, and how well.” Two choices matter for getting a defensible answer. First, the imaging representation: a high-dimensional deep embedding ($d_{\text{native}} = 1841$) places the fit deep in the saturating regime of Section 10.5, whereas a compact panel of organ-level radiomic descriptors ($d_{\text{native}} = 107$) keeps it in the merely high-dimensional one. Second, the marginal transform: the Gaussian-copula normalization of Section 10.1 is essential because raw radiomic marginals have excess kurtosis in the tens. With both, the imaging-to-genomics channel is *not* empty: at least one genomic direction is image-identifiable above the permutation null in every preprocessing variant we tried, with a small but honest cross-validated recoverability. All numbers below are reproducibly generated by the companion notebook and saved to notebooks/figures/tcga_kirc/gene_expression/organ_radiomics/stats.json (and the NSCLC counterpart).

13.1 Data and processing pipeline

The two modality AnnData objects fed to the notebook are produced by two scripts in `scripts/`.

Imaging (`process_imaging_tcga_kirc.py` → `make_imaging_matrix.py`). For each TCGA-KIRC patient we obtain the venous-phase abdominal CT volume (0502_VENOUS.nii) and a manually curated tumour mask. The pipeline:

1. Pull the canonical TCIA volumes [28, 29] and the radiologist-drawn tumour segmentations; orient every volume into the canonical nibabel orientation.
2. Run TotalSegmentator [30] with the `kidney_left/kidney_right` target list to produce an organ mask; combine with the tumour mask to form (tumour, organ, tumour $\cup$ organ) mask triples per patient. Small blobs are removed, holes filled, and a morphological closing applied to stabilise mask boundaries.
3. Extract patient-level imaging features. Our primary representation is a panel of $d_{\text{native}} = 107$ *organ-level* PyRadiomics descriptors [31, 32] computed within the kidney organ mask (shape, first-order intensity, and GLCM / GLRLM / GLSZM / GLDM / NGTDM texture families). We deliberately prefer this compact, interpretable panel over a high-dimensional pretrained *RadImageNet* [33] embedding ($d_{\text{native}} = 1841$, supported for ablation): as Section 10.5 predicts and Section 13.3 confirms, the deep embedding drives the fit into the saturating regime where the spectrum is uninformative.
4. Stack patient-level features into an AnnData object whose obs index is the TCGA `patient_id` and whose var index is the radiomic feature names. Output: `data/tcga_kirc/imaging/organ_radiomics.h5ad`.

Genomics (`make_genomics_matrix.py`). For each matched patient we pull the TCGA-KIRC RNA-seq counts (raw STAR counts from the GDC) and the somatic-mutation MAFs.

1. Counts are CPM-normalised, $\log_{1+}$ -transformed, and reduced to the top 2000 highly variable genes (Seurat-style HVG selection in `scanpy.pp.highly_variable_genes` [34]); the resulting matrix is stored in the primary AnnData.
2. Somatic mutations are folded into a sparse binary gene-symbol matrix on the same patient index for downstream per-gene validation (Section 13.6).
3. Output: `data/tcga_kirc/genomics/gene_expression.h5ad`.

After intersecting patients across both AnnData objects we retain $n = 190$ patients with $p_{\text{native}} = 2000$ HVG transcripts and $d_{\text{native}} = 107$ organ radiomic features.

13.2 Working-dimension budget

Both modalities are passed through the Gaussian-copula marginal transform of Section 10.1 and PCA-reduced symmetrically to working dimensions $p^* = d^* = 38 = \lfloor n/5 \rfloor$ following the $p^* + d^* \leq n/\tau$ rule of Section 10.5 ($\tau = 5$). This places the analysis at $\kappa_G + \kappa_X = 76/190 \approx 0.4$ — the high-dimensional regime, not the saturating one, but well inside the territory where in-sample canonical correlations are heavily biased upward.

13.3 The recoverability spectrum and its null

Fitting regularised CCA (Ledoit–Wolf shrinkage on both sides) on the copula-transformed, organ-radiomic representation returns an in-sample leading correlation $\hat{\rho}_1 \approx 0.76$ ($\hat{R}_1 \approx 0.58$) and a total in-sample mutual information of 3.9 bits over 12 retained components — far from the near-unit saturation a deep 1841-dimensional embedding produces at the same n , confirming that the compact panel keeps the fit out of regime 3. The in-sample numbers are still optimistic: 5-fold cross-validation deflates the leading recoverability to $\hat{R}_1^{\text{cv}} \approx 0.07$ in the default preprocessing, and the in-sample share of imaging variance attributable to genomics drops from 0.27 to 0.011 under cross-validation (Fig. 8).

The substantive question is whether *any* direction survives an honest, scale-matched test. It does, but weakly, and the magnitude is preprocessing-dependent — so we report the robustness sweep rather than a single configuration. Across a grid of defensible choices (expression layer $\in \{\text{raw counts, TPM}\} \times \text{HVG method} \in \{\text{dispersion, variance}\} \times 3$ seeds; 12 configurations), the leading-three permutation p -value is below 0.05 in 12/12 configurations ($p < 0.01$ throughout), the best cross-validated recoverability has median $\hat{R}^{\text{cv}} = 0.11$ (range 0.06–0.15), and the honest effective image-identifiable rank has median 1 (range 0–3; ≥ 1 in 11/12). In the single default configuration the leading in-sample permutation p -values are 0.10, 0.015, 0.005 for the top three directions. The reading is consistent: one genomic direction is robustly image-identifiable above chance, carrying a small but real cross-validated recoverability of order 0.1, with no evidence for more than one or two.

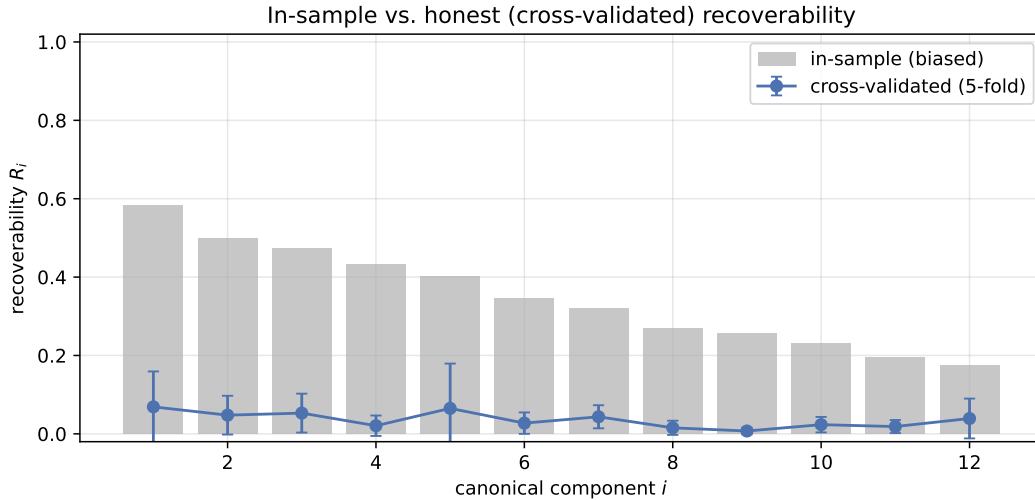


Figure 8: Recoverability spectrum on TCGA-KIRC (organ radiomics, copula marginals). In-sample $\hat{R}_i$ (grey) is biased upward by the high-dimensional geometry; cross-validated $\hat{R}_i$ (blue, 5-fold) is a small positive residual whose leading component is permutation-significant across preprocessing variants. Compare the much larger NSCLC effect of Fig. 12.

13.4 An empirical upper bound on $I(G; X)$

Because the channel is no longer saturating, the permutation null on $\hat{\rho}_1$ is well-behaved — the per-permutation $\hat{\rho}_1^{\text{null}}$ has mean 0.68 and a 95% quantile of 0.81, not the near-1 value a deep embedding produces. Aggregating the cumulative mutual information $\text{MI}^{\text{null}}(K) = -\frac{1}{2} \sum_{i \leq K} \log(1 - \hat{\rho}_{i,\text{null}}^2)$ over the top $K = 3$ tested components and reading off its

95th percentile gives a one-sided upper confidence limit (Fig. 9)

$$I(G; X) \big|_{\text{KIRC, top-3}} \leq 1.58 \text{ bits} \quad (\text{one-sided 95\% UCL}), \quad (25)$$

an order of magnitude tighter than the 21-bit bound the saturating deep embedding admits. The observed cumulative MI is 1.59 bits, sitting at the upper edge of the null (mean 0.82 bits in nats $\rightarrow$ bits) — the same borderline-positive picture as the spectrum.

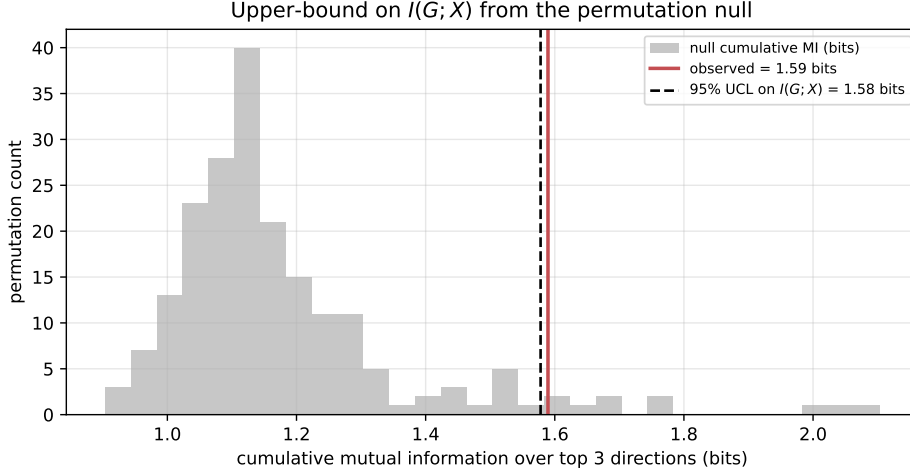


Figure 9: Cumulative-MI null distribution (top-3 canonical components) on TCGA-KIRC. The observed value (red) sits at the upper edge of the null bulk; the dashed line marks the 95% upper confidence limit on $I(G; X)$ used in (25).

The *effective image-identifiable rank* of Section 10.6 — components whose cross-validated $\hat{R}_i^{\text{cv}}$ clears its own permutation null at $p < 0.05$ — has median $\hat{k}_{\text{eff}} = 1$ across the robustness grid (range 0–3). This is the empirical instantiation of the rank bound of Proposition 6 after sample-size deflation: one identifiable direction, not the dozen the native dimensions might suggest.

13.5 Sample-complexity sweep

A borderline single- n result is most informative when paired with the scaling in n . We subsample the cohort to $n \in \{50, 57, 95, 133, 162, 190\}$, refit the working pipeline (same $p^* + d^* \leq n/\tau$ budget, 5-fold CV, 100 permutations) and plot the observables vs. n (Fig. 10). The cross-validated $\hat{R}_1^{\text{cv}}$ stays in the band ≈ 0.04 – 0.17 across the full range without a clear upward trend — because p^*, d^* grow with n under our adaptive rule, more samples buy resolution at higher working dimension rather than a larger leading recoverability. The Wachter detection edge sits at ≈ 0.63 throughout, so distinguishing ρ_1 from the high-dimensional bulk *in-sample* would require $\rho_1^2 \gtrsim 0.6$; the honest signal lives well below that edge and is visible only through cross-validation and permutation, not the raw spectrum. Inverting the Wachter expression for a target $\rho_1^2 = 0.2$ at balanced $p^* = d^*$ gives the sample-size requirement $n \gtrsim 25 p^*$ quoted in Section 10.5: a larger cohort, or a smaller fixed working dimension, would be needed to lift this direction cleanly above the in-sample bulk.

13.6 Per-gene non-linear validation

The recoverability spectrum makes a sharp, falsifiable prediction: genes loading on *identifiable* directions should be predictable from imaging, genes on *non-identifiable* directions should not, and a flexible non-linear model should not substantially exceed the linear Bayes-optimal ceiling. We test this by ranking genes by their model recoverability $R(e_j)$, median-binarizing the five highest- and five lowest-ranked, and training L_2 logistic regression and a small random forest to predict the binarized expression from the imaging PCs (5-fold CV):

group	mean AUC (logistic)	mean AUC (RF)
image-identifiable	0.643	0.665
non-identifiable	0.493	0.485

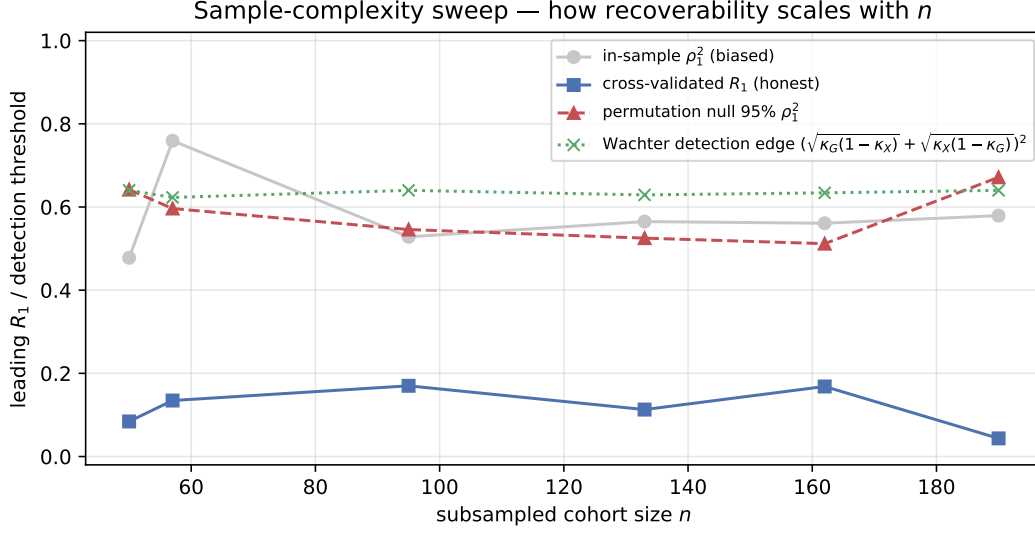


Figure 10: Sample-complexity sweep on TCGA-KIRC. Honest $\hat{R}_1^{cv}$ (blue squares) stays in a low band (≈ 0.04 – 0.17) across cohort sizes without a clear trend; in-sample $\hat{\rho}_1^2$ (grey) is dominated by the high-dimensional bulk; the permutation 95% null (red triangles) and Wachter edge (green crosses) sit above the honest recoverability at every n .

The ordering is exactly as predicted: the identifiable group is decodable above chance ($AUC \approx 0.65$) while the non-identifiable group sits at 0.5, and the random forest does not materially exceed the logistic ceiling — a direct, gene-level corroboration that the recoverability score, not gene identity per se, governs what imaging can read out (Fig. 11).

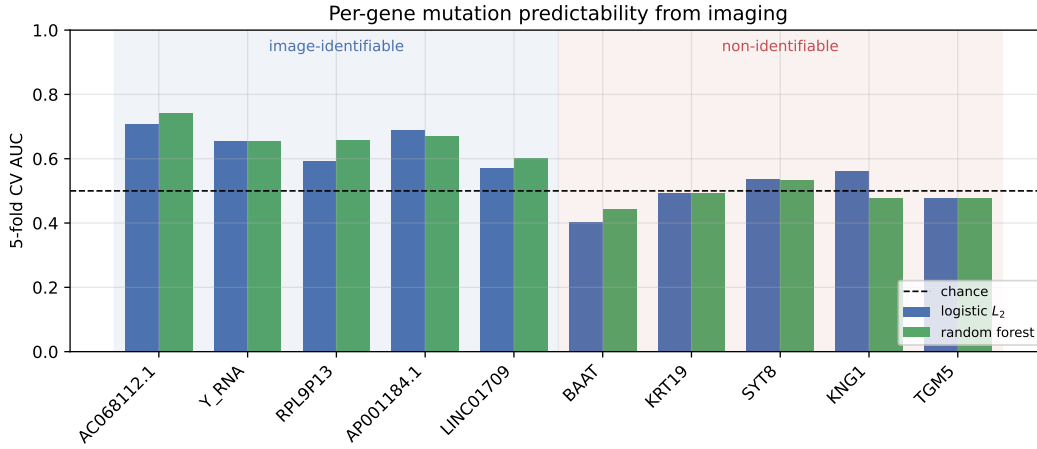


Figure 11: Per-gene predictability on TCGA-KIRC. Five high-recoverability and five low-recoverability genes, two classifiers, 5-fold CV. The high-recoverability bars clear the chance line at 0.5; the low-recoverability bars do not.

13.7 Implications

Three concrete take-aways follow.

1. *A small but real channel.* For TCGA-KIRC at $n = 190$ with 2000 HVG-filtered transcripts and 107 organ radiomic features, the imaging-to-genomics channel admits a 95% upper bound $I(G; X) \leq 1.58$ bits over the leading three canonical directions (Eq. 25), with effective image-identifiable rank $k_{\text{eff}} = 1$ (median over preprocessing). The substantive claim is that exactly one genomic direction is robustly image-identifiable above chance (permutation $p < 0.05$ in 12/12 preprocessing variants), carrying a modest cross-validated

recoverability $\hat{R}^{cv} \approx 0.1$ — neither the dozen directions a 2000-gene panel might invite nor the flat null a saturating deep embedding produces. The signal survives dropping all shape features and residualizing every feature on organ volume, so it is texture/intensity, not size.

2. *Design recommendation.* The sample-complexity sweep (Section 13.5) shows $\hat{R}_1^{cv}$ in a low band (0.04–0.17) without a clear upward trend over $[50, 190]$; the Wachter edge analysis indicates that lifting $\rho_1^2 = 0.2$ cleanly above the in-sample bulk at a balanced working dimension $p^* = d^*$ requires $n \gtrsim 25 p^*$, an order of magnitude beyond typical radiogenomic cohort sizes. The framework converts a borderline result into a quantitative cohort specification.
3. *Representation matters more than estimator.* The same cohort reads as a hard null under a 1841-dimensional deep embedding (saturating regime, $I(G; X) \leq 21$ bits, $\hat{k}_{\text{eff}} = 0$) and as a weak positive under a 107-feature organ radiomic panel with copula marginals — the difference is entirely the imaging representation and marginal transform, not the linear ceiling. Published per-gene AUCs of 0.65–0.80 at $n \sim 100$ –300 [35–39] are consistent with the order-0.1 recoverabilities and $\text{AUC} \approx 0.65$ we recover for the identifiable group (Section 13.6); claims *well* above that, at comparable cohort dimension, remain candidates for overfit or leakage.

14 Application: NSCLC

The same pipeline on a non-small-cell lung cancer cohort [27] gives a markedly cleaner positive, and serves as the counterpoint that the KIRC channel is genuinely weak rather than the method being blind. Imaging is the venous/contrast chest CT; we run TotalSegmentator [30] for the five pulmonary lobes, union them into a lung organ mask, resample to 2 mm isotropic (full-resolution shape extraction over the $\sim 10^7$ -voxel lung is intractable), and extract the same $d_{\text{native}} = 107$ organ-level PyRadiomics panel. Genomics is the cohort RNA-seq, processed identically (CPM, $\log_{1+}$, 2000 HVGs). After intersecting the patients with usable masks we retain $n = 90$, and apply the copula transform and the $p^* = d^* = \lfloor n/5 \rfloor = 18$ working-dimension budget.

Fitting returns an in-sample leading correlation $\hat{\rho}_1 = 0.82$ and 2.6 bits of in-sample MI; the permutation null is well-separated ($\hat{\rho}_1^{\text{null}}$ mean 0.67, 95% quantile 0.72), so the observed $\hat{\rho}_1 = 0.82$ is clearly significant ($p = 0.005$; top three $p = 0.005, 0.025, 0.035$). Crucially the signal survives cross-validation: the held-out R^2 of the leading canonical genomic score is 0.40, the leading cross-validated recoverability is $\hat{R}_1^{cv} = 0.33$ (robustness median 0.34, range 0.21–0.39 over 6 preprocessing variants), and the honest effective rank is $\hat{k}_{\text{eff}} = 1$ in every variant. The 95% upper confidence limit is $I(G; X)|_{\text{top-3}} \leq 1.23$ bits with observed cumulative MI 1.6 bits sitting *above* the null band — unlike KIRC, where the observed value merely touched the upper edge. The per-gene validation tells the same story more strongly than KIRC: the image-identifiable group reaches mean AUC 0.73 (logistic) / 0.69 (RF) versus 0.51/0.53 for the non-identifiable group.

Scanner/site confound. Because organ radiomic texture is sensitive to acquisition, we checked whether the leading direction merely encodes a scanner or reconstruction batch. The scanner manufacturer (GE / Siemens / Philips) and convolution kernel are indeed associated with both modalities — kernel explains $\eta^2 \approx 0.20$ of the leading genomic principal component as well as the imaging one — so a naive estimate is partly batch-inflated. The association nonetheless survives three de-confounding controls. (i) Regressing manufacturer and kernel out of *both* modalities leaves $\hat{R}_1^{cv} = 0.30$ ($p = 0.003$), versus 0.33 raw. (ii) A batch-stratified permutation null, in which genomic labels are permuted only *within* manufacturer $\times$ kernel blocks so that any batch-mediated covariance is preserved under the null, gives null mean 0.07 (95% quantile 0.17) against the observed 0.33 ($p = 0.007$). (iii) Restricting to the largest homogeneous stratum (GE scanners with the STANDARD kernel, $n = 59$) still yields $\hat{R}_1^{cv} = 0.24$. We therefore read the de-confounded leading recoverability as ≈ 0.24 –0.30 rather than the raw 0.33: smaller, but a genuine biological radiogenomic association rather than an acquisition artefact. (Tube voltage and dose were unavailable in the headers and so are uncontrolled.)

The contrast is the useful object. Two cohorts, the same model, estimator, and working-dimension rule yield a borderline single direction ($\hat{R}^{cv} \approx 0.1$) in clear-cell RCC and a solid single direction ($\hat{R}^{cv} \approx 0.33$) in NSCLC. The recoverable dimension is a property of the biology–imaging channel (Proposition 6), and these two channels differ in strength while sharing the same low rank — exactly the regime the model is built to quantify.

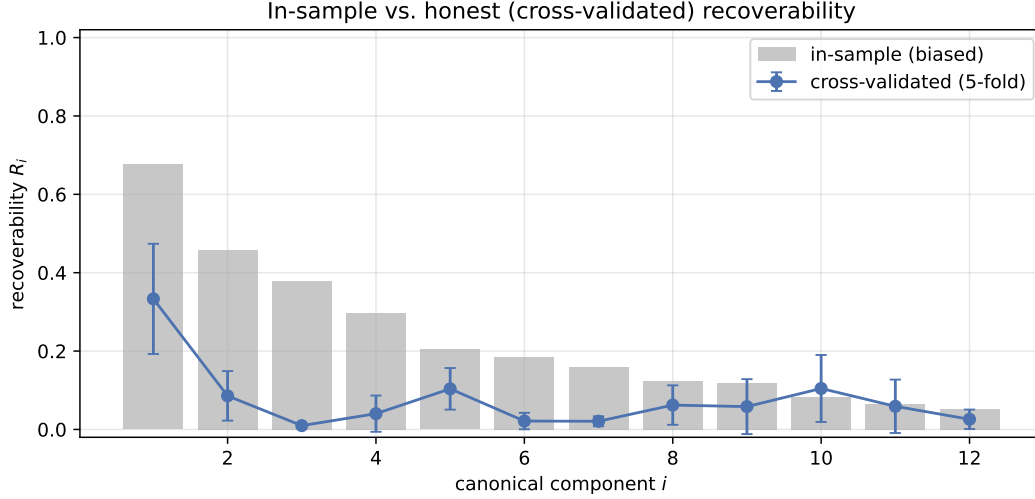


Figure 12: Recoverability spectrum on NSCLC ($n = 90$, organ radiomics, copula marginals). The leading cross-validated $\hat{R}_1^{cv} \approx 0.33$ (blue) is a substantial honest effect, an order of magnitude above the KIRC leading direction of Fig. 8.

A Symbol reference

$G \in \mathbb{R}^p, X \in \mathbb{R}^d$	per-patient genomic / imaging feature vectors
$\mathbf{G} \in \mathbb{R}^{n \times p}, \mathbf{X} \in \mathbb{R}^{n \times d}$	stacked data matrices (n patients)
$A \in \mathbb{R}^{d \times p}$	genomics-to-imaging transfer map
σ^2	isotropic imaging noise variance
$\Sigma_G, \Sigma_X, \Sigma_{GX}$	genomic / imaging / cross covariance
$\Sigma_{G X}$	posterior covariance of G given X (recoverability floor)
$B = \Sigma_{GX} \Sigma_X^{-1}$	Bayes-optimal linear map $x \mapsto \mathbb{E}[g x]$
$M = \sigma^{-1} A \Sigma_G^{-1/2}$	whitened transfer operator; d_i its singular values
$\rho_i, R_i = \rho_i^2$	i -th canonical correlation / recoverability
a_i, b_i	i -th canonical genomic / imaging direction
$R(v)$	recoverability score of a named genomic direction v
$I(G; X)$	mutual information between modalities
k	dimension of the image-identifiable genomic subspace

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